

MARSILI

*An Interactive Hybrid Game Experience for Heritage, Science, and
Discovery*

Project Brief

Bologna, Italy
Museum and Heritage Interactive Media Design

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1. Project Overview

Echoes of Marsili is an interactive hybrid game based on Luigi Ferdinando Marsili and his life, science work, and legacy in Bologna history. The project mainly tries to make the Marsili museum more interesting and easier for visitors to connect with. Instead of just looking at objects and reading information, visitors can interact with the story and take part in the experience. The idea is to make people feel more involved while learning about Marsili and his discoveries. In the future, the project also hopes to bring more attention to Marsili's heritage and improve the overall museum experience for visitors.

Marsili original collection and manuscript uses, presenting him as a guide. Each visitor explore the museum from his perspective, engaging with both physical artifacts and digital interactions for a unified and accessible experience.

Human centered interactive design principles guide all aspects of the experience, including navigation, information delivery, object interaction, and feedback. The aim is to make museum visits feel like real exploration.

2. Context and Background

2.1 The Marsili Collection in Bologna

Luigi Ferdinando Marsili's maps, manuscripts, scientific tools, and natural history objects are kept in museums in Bologna. These real objects are important for the project because they connect the story with Marsili's actual life and work. Visitors can learn about his discoveries in a more interesting and interactive way. Marsili worked in many different fields. He was a scientist, traveler, military officer, map maker, and collector. Because of this, his life includes many stories about travel, science, and the problems people faced when trying to understand the world in the seventeenth and eighteenth centuries.

2.2 The Museum as a Starting Point

One important idea from the course was that heritage design should start with the place and its history, not only with technology. This idea helped guide the project from the beginning. The museum space, objects, layout, and the way visitors move around became an important part of the design. Digital features are only added to support and improve the experience, not to take attention away from the real museum and its objects.

There is also a practical reason to prioritize the museum setting over digital screens. Visitors who choose to attend in person should be rewarded with meaningful activities that make their physical presence valuable.

3. The Design Problem

3.1 Passive Museum Engagement

In many museums, visitors just walk around the rooms, read small labels, see objects, and move to the next place. Most of the time there is not much interaction, so some people can feel bored or lose interest quickly. This is not a bad thing about museums, but it is how many traditional museums still work today. Because of this, visitors may leave without really feeling connected to the story or information they saw.

Echoes of Marsili tries to change this experience by making visitors more active. Visitors can interact with the story and take part in small activities inside the museum instead of only watching. This makes the visit more fun, personal and easier to remember.

3.2 Accessibility and Inclusivity

Another problem is that not every visitor understands museum information in the same way. If there is too much difficult text, only one language, or hard science topics, some people can feel left out. Families with kids, tourists, or visitors who never studied old science topics may not fully understand everything during the visit.

For this reason, the project tries to make the experience more simple and easy for everyone. Visitors do not need special knowledge about Marsili or strong technology skills to use it. The language, navigation, and design was made in a clear and easy way so different people can enjoy the experience without feeling confused.

3.3 Making Heritage Feel Alive

A big challenge in the project was making Marsili's world feel less old and more interesting for visitors. Many museum objects like maps, manuscripts, and science instruments can look difficult or confusing at first. Some visitors maybe don't understand why these objects are important, so they just look at them for a short time and continue walking.

To improve this, the project adds small interactions and storytelling so the objects feel less boring and more connected to the visitor. People can explore the story while moving through the museum, instead of only reading information boards. This makes the experience feel more real, engaging and easier for visitor to remember later.

4. Main Concept

The main idea of the project is that visitors do not only learn about Marsili, but also follow his journey and way of thinking. Marsili becomes part of the story, guiding visitors through his discoveries and experiences. Instead of only looking at museum displays, visitors slowly uncover his world by exploring the objects and stories he left behind.

This changes the way visitors connect with the museum content. Rather than just reading information, they interact with Marsili's ideas, observations, and discoveries. The experience feels more like exploration than a normal lesson. Visitors can scan, touch, and explore different parts of the museum, unlocking new stories and information during the visit. Some interactions may reveal a scientific instrument, an old manuscript, or a map connected to Marsili's travels and work. Through these small discoveries, visitors slowly build a better understanding of Marsili's life, his scientific work, and his importance in history.

5. Why Marsili Works as a Subject

Some historical people can be difficult to turn into interactive museum experiences because there is sometimes not enough material or their stories feel too abstract. Marsili is different because his life and work left behind many real objects and stories that visitors can explore. He worked in many areas like ocean studies, military engineering, map making, natural history collecting, and helping create scientific institutions. Because of this, visitors can discover many different sides of his life during the experience.

Marsili also lived a very adventurous life. He travelled to many places, was captured during war and later freed, studied the sea floor, and created early detailed maps of the Bosphorus. These real events make his story naturally interesting without needing to add fictional details. Another important thing is that Marsili's legacy still exists in Bologna museums today through manuscripts, instruments, maps, and specimens. Visitors can directly see these real objects, which helps make the interactive experience feel more real and connected to history.

6. Target Audience

The project is mostly made for young visitors between 14–27 years old who likes history, science, stories, and digital activities. Many people in this age group enjoy exploring and making choices instead of only reading long museum texts. Because of this, the project tries to make the museum visit feel more active and less boring than normal exhibitions. By using storytelling, museum objects, and small digital interactions together, visitors can feel more

connected with the experience.

Accessibility is also important in the project. The experience is designed for different visitors, including people with hearing or visual problems. Things like subtitles, audio help, simple text, and easy navigation makes the experience easier to use. By mixing sound, text, and visuals, more visitors can explore Marsili's story in their own way and without much difficulty.

7. Learning Goals

7.1 Understanding Marsili and His Heritage

One learning goal of the project is helping visitors know more about Marsili as a real person, not only a historical name. The experience shows how he travelled, explored places, collected objects, and studied the sea. Visitors can also learn about the scientific work he did and why he became important in Bologna's history. By the end of the visit, people should understand Marsili better and know why his work still matters today.

7.2 Understanding Scientific Inquiry and Knowledge Production

Another goal of the project is showing how scientific knowledge was created during Marsili's time. Science was not only done inside laboratories, but also during travelling, observing nature, collecting objects, and sharing ideas with other people. Through the interactive activities, visitors can see how curiosity and observation helped Marsili in his discoveries. The project does not only focus on teaching information, but also helping visitors understand how learning and discovery happens.

7.3 Heritage, Interpretation, and Curiosity

Another learning goal of the project is helping visitors think differently about museum objects and heritage. The experience shows that manuscripts and old objects are not just old things to look at, but also signs of ideas, discoveries, and human thinking. A manuscript, for example, is not only a old paper — it also shows what questions a person asked and how they tried to understand the world.

If visitors leave the museum feeling more curious and more interested in asking questions about objects and their meaning, then the project has achieved an important goal.

8. What the Game Actually Solves

8.1 Transforming Passive Observation into Active Participation

In many museums, people usually walk around, read small texts, look at objects, and then move on. After the visit, they may forget most of what they saw. This project tries to make visitors more active and involved. Visitors can listen to sounds, use their phones, make choices, and follow the story while walking through the museum. This makes the experience feel more interesting, personal, and easier to remember.

8.2 Improving Accessibility and Engagement

The project also tries to make the experience easier for many kinds of people. The text is kept more simple, navigation is easier to follow, and visitors can interact by touching and exploring. Different language options also helps visitors who may not understand everything in Italian. The idea is that more people should feel comfortable while using the experience, even if they are not very confident with technology.

8.3 Creating Meaningful Digital Interaction

Another issue in some museums is that digital technology can feel too complex or not connected with the real objects around the visitor. In this project, the digital parts are kept simple so they support the museum visit instead of taking all the attention. The technology works together with the objects and stories in the museum, so visitors still focus on the real heritage around them.

9. Hybrid Design Direction

In this project, “hybrid” means the experience happens between the real museum space and the digital story. Visitors use real museum objects to unlock parts of the story on their phones, and the digital activities also guide them to different places inside the museum. Both the physical and digital parts work together during the visit.

This is important because fully digital museum experiences can be used anywhere, so people may not feel a strong reason to visit the museum itself. With this hybrid system, visitors need to be inside the museum to fully experience the story and interaction. The project uses things like QR codes, touch interaction, object-based storytelling, and small digital activities to connect the museum space with the narrative. Everything is kept simple so visitors can use the experience naturally without needing to understand how the system works behind the scenes.

10. Digital Elements

10.1 Twine-Based Narrative Structure

Twine is used in the project to build the story and interactive parts. It helps connect different choices, scenes, and information together without needing to make a full game from zero.

Visitors can scan a QR code near an object, old manuscript, or museum place, and then open a part of the story on their phone. In this way, the digital part stays connected with the real museum space. It also makes small updates easier later without changing the physical setup inside the museum.

10.2 Cross-Device Compatibility

Visitors can open the experience on different devices like phones, tablets, or computers. This makes it more easier for museums because people can use their own device during the visit. Small sounds or animations are added in some parts, but the experience can still work without them. The main story and interaction stays simple so visitors can still understand what is happening.

11. Physical Elements and Spatial Logic

The project is connected closely with the real Marsili museum space. Things like old manuscripts, maps, science instruments, display boxes, and the museum rooms are all part of the experience. They are not only decoration for the digital story, but an important part of it.

The design also follows the way people normally walk and explore inside a museum. Visitors can follow visual signs, objects, and interaction points without needing too many instructions. Moving from one place to another becomes part of the story itself.

The project tries to keep the experience feeling natural and simple for visitors. The system works with the way people already move around the museum, so the interaction does not feel too complicated or forced.

12. Accessibility and Inclusion

12.1 Inclusive and Intuitive Interaction

The project tries to make the interaction feel simple from the beginning. Visitors should not need to read many instructions before using the experience. If people get confused too fast, then the interaction is probably not working well. Because of this, buttons, navigation, and

visual signs are made clear and easy to see. The main idea is that visitors spend more time exploring the museum content instead of trying to figure out how the system works.

12.2 Multilingual and Multi-Modal Communication

The experience is made for different kinds of visitors, including tourists, students, and people from different countries. Some visitors maybe do not understand academic Italian very well, so the project uses simple text, visuals, and more than one language. Important information is also shown in visual ways, not only through audio. This can help visitors who have hearing difficulties or people in noisy museum spaces where sound is harder to hear properly.

12.3 A Comfortable and Guided Experience

The project also tries to make visitors feel comfortable while moving through the museum. People should mostly know where to go and what to do next without feeling lost. If visitors become unsure, the system gives small hints or guidance in a natural way. The difficult part of the experience should come from thinking about the history and objects, not from struggling with the interface or controls.

13. Narrative Tone and Experience Feel

13.1 Curiosity, Discovery, and Immersion

The tone of the project is mostly based on curiosity and discovery. Marsili was a person who always wanted to learn new things and understand the world around him, and this feeling is part of the whole experience. Visitors are encouraged to keep exploring and see what comes next instead of feeling like they just finished one task and move on.

The project also uses immersion, but in a simple and careful way. It does not try to make visitors forget they are inside a museum. Instead, the feeling comes slowly through the story, the objects, and the way different parts of the visit connect together. The experience should feel natural and not too dramatic or artificial.

13.2 Respectful and Engaging Heritage Storytelling

The project talks about Marsili and his work in a respectful but still interesting way. The goal is to avoid a presentation that feels too formal, heavy, or difficult to follow. Instead of only showing facts and dates, the experience also tries to show Marsili as a real person with curiosity, ideas, and personal experiences.

13.3 Creating Emotional Connection

Another important part of the experience is helping visitors feel more connected to the history

and museum material. People usually remember history more when it feels personal or relatable to them. Because of this, the project shows moments from Marsili's life that include curiosity, travel, problems, discoveries, and ambition.

Museum objects, old documents, maps, and collected materials are not only decorations for the story. In many parts, they actually become the story itself. Each object can show something small about how Marsili worked, what interested him, or what he was trying to learn at that time.

14. Content Strategy

14.1 Heritage-Driven Narrative Content

The content strategy centers entirely on Marsili's actual legacy. Manuscripts, maps, scientific instruments, collected specimens, and archival materials from the museum collections serve as the primary narrative anchors. These materials aren't used merely as illustrations of the story—they are the story. Each object or document reveals something specific about how Marsili worked and what he was trying to understand.

14.2 Interactive and Meaningful Content Integration

Every piece of content within the experience is connected to an interaction and to narrative progression. There's no content that exists simply to be read without doing anything. Objects unlock stories. Documents reveal context. Maps connect to geographical explorations. The content is always embedded in the interaction structure rather than placed alongside it.

14.3 Layered Information Structure

The project gives information step by step. Visitors first see the main and simple information. If they want, they can also open more details about history or science later.

Not every visitor likes to read a lot, so the experience tries to work for different people. Some visitors maybe only look at the basic parts, while others spend more time exploring extra content. In this way, the system stays simple but still has enough information for people who are more curious.

15. Gameplay Structure and Interaction Flow

The experience is based more on exploration and discovering things little by little. Visitors are not only reading information, but also interacting with objects, scans, small story parts, and digital content while moving through the museum. The system tries to make people curious and continue exploring more.

The experience is also not competitive. There is no score, ranking, or “game over” moment. Visitors can move at their own speed and choose what interests them more. The idea is to make people feel comfortable, not tested like in a school exam.

Small activities and story moments help guide visitors through different parts of Marsili’s life and work. Sometimes a visitor may discover a new fact, while other times they may understand an object or old manuscript in a more personal way.

The gameplay structure can still change during development because the project is in a prototype stage. But the main idea stays the same: visitors explore museum materials, receive simple feedback from the system, and slowly build a better understanding of Marsili and his story through active interaction.

16. Prototype Direction

16.1 Iterative Prototyping Process

The project is developed little by little through testing and changes during the process. Instead of making a complete final version from the start, the team first creates simple prototypes to check what works better and what maybe needs improvement. This helps the team test the story flow, interaction, usability, and overall visitor experience before making final decisions later.

16.2 Tools and Formats

Different tools are used during the prototype stage. Twine is used for the interactive story and different story paths. Figma is used to make interface ideas and simple screen layouts. The team also uses paper prototypes to test interaction ideas and movement in the museum space before building everything digitally.

These tools help the team experiment more freely and change things faster when needed. Some parts maybe work well in testing, while other parts may need improvements later.

16.3 Purpose of Prototyping

The prototype is not the final version of the project. It is mainly used to test ideas,

interactions, and storytelling at an early stage. It also helps the team notice problems early before spending too much time building the final system.

Feedback from users is very important during this process. It helps the team understand what parts feel clear, confusing, interesting, or difficult for visitors. Because of this, the design may still change and improve many times development.

17. Project Values and Success Criteria

The project is based on four main values: accessibility, curiosity, participation, and meaningful connection with cultural heritage. These ideas help guide many of the design choices during the development process.

The success of the project is not only about technology or visual design. What matters more is the visitor experience. The experience should feel simple, interesting, and comfortable for different kinds of visitors and age groups. Visitors should leave with a better understanding of Marsili and feel that they were part of the discovery process, not only reading information.

If the digital parts and physical museum space feel separate or confusing, then the project is not fully successful, even if some individual features work properly on their own.

18. Interaction Design Principles

The interaction design follows three main ideas: clarity, simplicity, and easy navigation. These ideas are used in all parts of the experience, from the main navigation to small interface details.

Each interaction is made to feel clear and direct. When visitors scan, touch, choose, or explore something, the system gives a quick and understandable response. Visitors should easily know if something worked, what new information they found, or where they can go next.

The project also follows a user-centered approach. The experience is made to feel more like a journey through the museum, not just a group of different screens. Visitors move from one part to another in a simple and natural way. The project also thinks about how people start the experience, how they get information, and how they feel when the visit finish. Interaction is not used only for technology, but also to help the story feel more real and connected for visitors.

19. User Journey and Narrative Flow

19.1 Opening and Orientation

At the start, visitors are introduced to the experience in a simple and welcoming way. The beginning is not made to give too much information at one time, because that can feel confusing for some people. Instead, the opening gives small pieces of context and lets visitors slowly become curious about Marsili and the museum space. The idea is to help people feel comfortable first, then interested to continue exploring more.

19.2 Progressive Discovery

As visitors move around the museum, they slowly discover more parts of Marsili's life and work. Some information comes from objects, some from manuscripts, and some from digital interactions or small activities. Different visitors may explore in different ways or spend more time in some parts than others, and that is completely fine. The experience is made to feel flexible, so people can follow what interests them without feeling lost during the visit.

19.3 Closure and Reflection

Near the end of the experience, visitors get a little time to think about what they saw and explored in the museum. The ending does not finish too fast or suddenly. Instead, it connects different parts of the visit together and reminds visitors why Marsili's work is still important today. The idea is that people leave feeling they were part of the discovery, not only reading information from outside.

20. Content Inventory and Star Assets

20.1 Historical and Museum Assets

The project is built around historically significant materials connected to Luigi Ferdinando Marsili and the scientific heritage of Bologna. Core content assets include manuscripts, scientific instruments, cartographic drawings and maps, collected natural history specimens, archival materials, and museum display environments associated with Marsili's research and institutional contributions.

20.2 Narrative and Interactive Assets

Beyond the physical heritage materials, the project treats narrative and interaction as primary assets in their own right. Marsili's narrative presence functions as a guiding element throughout the experience. Interactive maps, scientific observation mechanics, and object-linked storytelling strengthen the project's identity by transforming historical materials into active components of the visitor journey.

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20.4 Experiential Value and Project Identity

These elements are considered star assets because they define the project's uniqueness, atmosphere, and educational value. Their integration into the interaction and narrative structure transforms heritage from a passive exhibition into a participatory experience of scientific research and cultural discovery.

21. Accessibility, Language, and Usability Requirements

21.1 Accessibility as a Core Requirement

From the start, the project tried to make the experience easy for many different people. Some visitors may be young, some older, and some maybe not used to digital things. Because of this, the project does not use very difficult interaction or confusing systems. The idea was simple: visitors should not feel stressed or lost while using the experience.

21.2 Language and Communication Design

The writing in the project is kept short and easy to read. Very academic or complicated words are not used too much because many visitors may not know them. Since people from other countries can also visit the museum, the experience includes more than one language. Later on, more translations can also be added if needed.

21.3 Usability and Multi-Modal Interaction

The project tries to make the museum experience feel natural and simple. Visitors can read text, look at visuals, touch parts of the interaction, and follow small instructions inside the museum space. Audio is optional, because some visitors maybe do not want to use headphones or cannot hear properly. Different ways of interaction are included so more people can enjoy the experience in the way that feels comfortable for them.

22. Team Roles and Collaboration

22.1 Collaborative Design Approach

This project was not made by only one person. Different team members worked on research, story writing, design, and digital interaction. Everyone shared ideas and helped during the process. Sometimes small changes were also made while working so the final experience could feel more connected and clear.

22.2 Distribution of Roles and Responsibilities

Each person in the team worked on different tasks depending on their skills and interests. Some people focused more on museum research and writing, while others worked on visuals, interaction ideas, or technical setup. Dividing the work like this helped the project move more easily and avoided too much pressure on one person.

22.3 Coordination and Shared Development

The team needed regular communication during the project. Members shared progress, discussed problems, and changed some plans when things were not working properly. Even with different roles, everyone worked together to keep the story, interaction, and design connected in one complete experience.

23. Constraints and Risks

23.1 Technical and Development Constraints

One challenge in the project is connecting digital interaction with the real museum space in a smooth way. The project also needs to stay possible within the available time, resources, and technical skills. Because of this, some features and ideas need to be carefully selected so the final experience stays clear and manageable during development.

23.2 Usability and Accessibility Risks

The experience is designed for many different people, including visitors who are not very confident with technology. If the interaction becomes too difficult or the navigation is unclear, some visitors may feel confused during the visit. To avoid this, the project uses simpler interaction, readable text, and easier navigation. Visitors should be able to move through the experience without too much effort while still feeling involved in the story.

23.3 Content Scope and Narrative Balance

Marsili's life and work include a lot of history, science, and cultural information. Adding too much content could make the experience feel heavy or difficult to follow. For this reason, the

project focuses only on some selected parts of his story that are more useful for storytelling and interaction. This helps visitors stay interested without feeling overloaded with information.

23.4 Meaningful Use of Digital Interaction

The digital parts of the project should always have a real purpose. QR codes, sounds, animations, and small interactions are only used when they help visitors understand the museum story better. The technology is not there to replace the museum objects, but to support the experience and make visitors feel more connected to the heritage around them.

24. Evaluation Criteria

24.1 User Experience and Interaction

The project is judged by how easy and clear the experience feels for visitors. Important things include if visitors can understand the interaction easily, if the experience is accessible, and if people stay interested during the visit. It also looks at how well the project explains Marsili's story and if the digital and physical parts work nicely together inside the museum.

24.2 Engagement and Educational Value

Another important point is if the project can make visitors feel curious and more connected with the museum content. The experience should make people explore, interact, and learn in a more active way. It also tries to show how storytelling, museum objects, and digital interaction can work together in one experience.

25. Theoretical Grounding

25.1 Design Rooted in Context and Heritage

The project is based on the idea that interaction design should come from a real place and real situation, not only from technology. Because of this, the Marsili museum, its objects, manuscripts, and historical rooms became the main base for the experience. The project is not trying to make a random digital app. Instead, it creates an interactive system connected closely to the museum and to how people visit cultural places today.

25.2 From Problem Identification to Design Requirements

The project started by looking at problems and visitor needs first. One important problem was that many museum visits can feel passive, where people only walk, read, and leave after short time. To answer this, the project focuses more on interaction, participation, and discovery. Things like simple navigation, language support, hybrid interaction, and clear guidance were added because they help visitors understand and use the experience more easily.

25.3 Interaction Design, Prototyping, and Interdisciplinary Practice

Another important idea behind the project is that interaction design is not only about visuals or making things look good. It is also about usability, behaviour, and visitor experience. During development, tools like Twine and Figma are used for testing story flow, interaction ideas, and user reactions before building the final version. The project also follows an iterative process, where ideas can change and improve during testing and collaboration between different team members.

25.4 Hybrid Museum Experience and Active Participation

The project also follows the idea of hybrid museum experiences, where visitors become more active instead of only observing things quietly. Marsili's life and scientific work are presented through interaction and exploration, not only through static information panels. Visitors are encouraged to take part, discover things slowly, and connect with the heritage material in a more personal way.

26. Summary

Echoes of Marsili is about making the museum visit more active and less repetitive for people. The project uses both real museum objects and simple digital interaction during the visit. Visitors do not only stand and read information from walls. They can scan things, explore small story parts, and move through the experience in their own way.

The project was made around Marsili's life, his research, and the objects connected to him in the museum. Some visitors may already know history very well, but others maybe never heard about Marsili before. Because of that, the experience tries to explain things in a more simple and easy way without making visitors feel lost or confused.

Another important part of the project is letting people interact at their own speed. Some visitors may spend more time reading or exploring, while others may only interact with a few parts. Both ways are okay. The design is kept clear and not too complicated so different kinds of visitors can still enjoy the experience.

In the prototype stage, tools like Twine, Figma, and paper mockups are used for testing ideas and interaction flow. The team can see what works good, what feels unclear, and what should change later. In the future, the final version could include QR codes, touch interaction, animations, and support for different devices inside the museum. Overall, the project tries to connect storytelling, museum heritage, and interaction in a way that feels more natural and personal for visitors.